

MLA02 - Fundamentals of Machine Learning

AT1 - Detailed Study Report

Question 1: House Price Prediction (Linear Regression)

A dataset of house sizes (sq. ft.) and prices was used. The data was checked for missing values, scaled, and split into 80% training and 20% testing samples. A Linear Regression model was trained on the training data and used to predict prices for the test data.

Results: R2 Score = 0.8608, MAE = 19316.31, RMSE = 23710.03.

The R2 score of 0.86 shows that house size explains about 86% of the variation in price. The predicted values closely follow the actual values, with a linear relationship between size and price.

Question 2: Student Performance Prediction (Multiple Linear Regression)

A dataset with four features - study hours, attendance percentage, internal marks, and assignment score - was used to predict final exam marks. The data was checked for missing values, scaled, and split into 80% training and 20% testing samples. A Multiple Linear Regression model was trained and used to predict marks for the test data.

Results: R2 Score = 0.9208, MAE = 3.66, RMSE = 4.41.

The R2 score of 0.92 shows that the four features together explain about 92% of the variation in final marks. Internal marks and study hours had the strongest influence on the predicted result, followed by attendance and assignment score.

Conclusion

The Multiple Linear Regression model (Q2) performed better than the Simple Linear Regression model (Q1), showing that using more relevant features improves prediction accuracy. Both models followed the same process: data loading, preprocessing, train-test split, training, prediction, and evaluation using the R2 score.